EDGE-GSE: SELECTIVE AUTOREGRESSION FOR LOW-LATENCY STREAMING GENERATIVE SPEECH ENHANCEMENT

Xin Yuan[1,*], *Junling Lv*[1,*], *Xingjun Tan*[1], *Rong Liu*[1], *Zezhou Xu*[2], *and Yanqiang Lei*[1,†]

[1] Guangzhou Shiyuan Electronic Technology Company Limited, Guangzhou, China
[2] Shanghai University of Finance and Economics, Shanghai, China
{yuanxin14959, lvjunling, tanxingjun, liurong, leiyanqiang}@cvte.com; xuzezhou@163.sufe.edu.cn

ABSTRACT

Autoregression need not be all-or-nothing in streaming codec-based speech enhancement. Temporal target-token feedback can propagate errors across frames, whereas fully non-autoregressive decoding ignores the ordered coarse-to-fine dependencies of residual vector quantization. We introduce Edge-GSE, a bounded-context, frame-wise generator that predicts a semantically supervised first codebook without temporal target-token feedback, generates the remaining codebooks autoregressively within each frame, and carries cross-frame state only through residual tokens. On URGENT 2025, a frame-aligned matched comparison shows that removing temporal feedback reduces bilingual macro ASR error by 45.0% relative. A double-blind listening test favors depth-AR over parallel residual decoding despite nearly identical DNSMOS. At the primary operating point, the complete system reaches an RTF of 0.81 on MTK 8678, corresponding to 208.3-ms context-plus-processing latency. Together, these results support selective rather than global autoregression for streaming codec-based speech enhancement. An on-device MTK demo and comparative audio samples are available at `https://delaygse.github.io/edge-gse/`.

Index Terms— streaming generative speech enhancement, neural audio codecs, residual vector quantization, selective autoregression, on-device inference

1. INTRODUCTION

Speech enhancement supports teleconferencing and mobile communication, hearing devices, in-vehicle voice interfaces, and noise-robust automatic speech recognition. These applications require suppressing noise and reverberation without sacrificing intelligibility, speaker identity, or responsiveness. Classical discriminative systems directly regress clean spectra or estimate time–frequency masks from noisy observations [1, 2, 3]. Their one-pass mappings are efficient and streamable, but under severe or unseen corruption a point estimate can over-smooth speech or suppress uncertain components instead of modeling plausible clean alternatives.

Generative speech enhancement addresses this ambiguity through two broad routes. Continuous-representation methods synthesize spectral or latent features with diffusion, flow matching, or Schrödinger bridges [4, 5, 6]. Discrete- representation methods quantize speech and generate neural-codec tokens with sequence or language models [7, 8, 9]. Both routes introduce a clean-speech prior to recover acoustic detail, yet many high-fidelity systems use whole-utterance, noncausal inference. Continuous generators often require repeated network evaluations, while token generators decode long sequences autoregressively; offline quality therefore does not ensure interactive deployment.

Streaming enhancement must emit frames before an utterance ends, under bounded future context and per-frame compute. Representative approaches include Stream.FM's frame-causal, few-step flow matching [10] and a low-latency codec model that autoregressively generates aligned clean tokens from past and current noisy frames [11]. The former still incurs numerical integration; the latter feeds generated tokens back across time, giving early errors a path into later predictions. Token-level NAR alternatives use parallel prediction or independent group quantizers [12, 13], but do not establish how a causal RVQ streamer should retain ordered residual dependencies. Genhancer's time-parallel configuration retains quantizer order but uses full-utterance noncausal conditioning [8]. Comparisons of global AR and NAR strategies expose this trade-off [14]; what remains unresolved is where autoregression should act in a bounded-context RVQ generator.

Our key observation is that the two RVQ axes require different feedback. Across time, generated-token feedback can propagate errors under bounded acoustic evidence; within a frame, depth conditioning preserves coarse-to-fine residual order without input look-ahead. In Edge-GSE (Fig. 1), w2v-BERT supervision designates CB_1 for content; a time-NAR predictor estimates CB_1 from bounded degraded speech, a depth-AR decoder generates CB_2–CB_{10} within the same physical frame, and a GRU routes cross-frame state only

*Xin Yuan and Junling Lv contributed equally to this work. †Yanqiang Lei is the corresponding author.

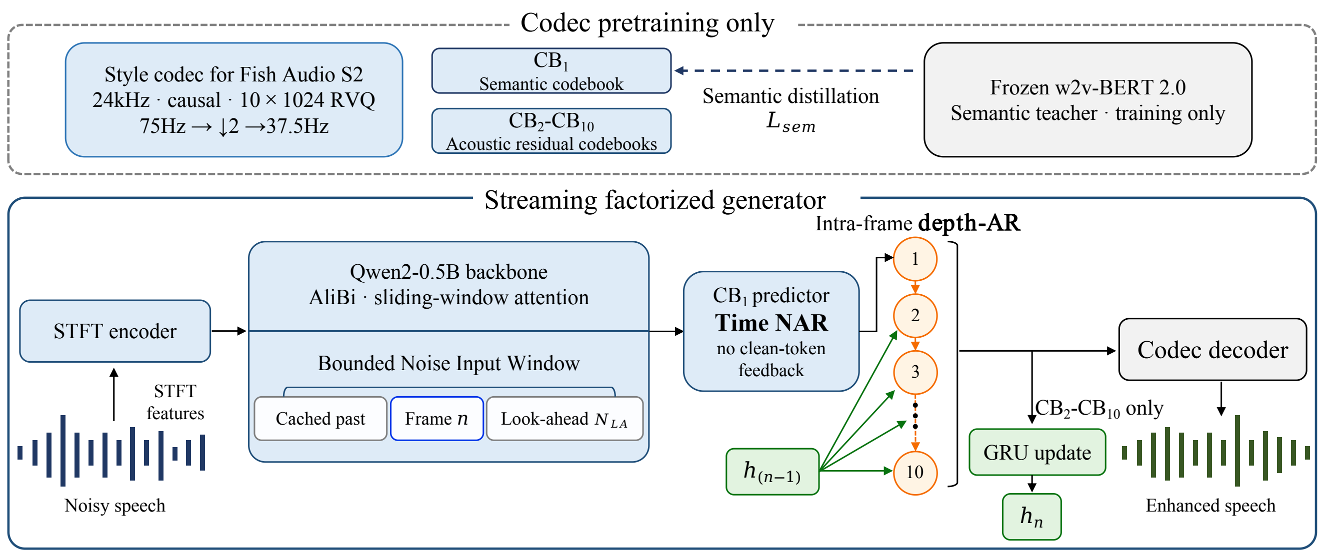

Fig. 1. Overview of Edge-GSE. Codec pretraining distills w2v-BERT 2.0 semantics into CB_1. At inference, time-NAR predicts CB_1 from bounded degraded-speech context without generated-token feedback; depth-AR then generates CB_2–CB_{10} sequentially within each codec frame, while the GRU carries cross-frame acoustic history only through the residual branch.

through residual tokens.

Matched controls on 400 URGENT 2025 utterances [15] support this asymmetric design. Removing temporal feedback reduces equal-language macro ASR error from 48.40% to 26.60% (45.0% relative). With temporal prediction fixed, a double-blind test records 103 judgments favoring depth-AR versus 15 favoring parallel residual decoding, with 42 ties, despite nearly identical DNSMOS. The complete system obtains 3.62 DNSMOS, 25.45% Macro ASR, and 0.708 SIM at 186.7-ms input context; its W8A16 C++ implementation reaches RTF 0.81 on MTK 8678, corresponding to 208.3-ms context-plus-average-processing latency.

Our contributions are:

- We formulate axis-specific feedback for streaming RVQ enhancement: content-designated time-NAR, intra-frame depth-AR, and residual-only cross-frame state.
- Matched objective and listening comparisons separately diagnose the effects of temporal feedback and RVQ-depth conditioning.
- Common-manifest evaluation, a 26.67–240-ms context sweep, and an on-device profile characterize the quality–latency trade-off.

2. PROPOSED METHOD

2.1. Selective Streaming Factorization

Edge-GSE separates the two dependency axes of an RVQ stream: the first codebook is predicted without temporal target-token feedback, whereas the remaining codebooks retain ordered within-frame prediction and residual-only cross-frame state. Let x and y denote a degraded waveform and its time-aligned clean target. At 37.5 frames/s, the pretrained codec represents clean frame n as $\mathbf{z}_n = (z_n^{(1)}, \ldots, z_n^{(K)})$ with $K = 10$ and frame duration $\Delta = 26.67$ ms. A causal front end maps bounded context from x to $\mathbf{s}_n$.

Let $\mathbf{c}_n = C_\psi(\mathbf{z}_{<n}^{(2:K)})$ summarize only residual tokens from preceding frames and let $\mathbf{r}_n = \mathbf{s}_n + \mathbf{c}_n$. For $\mathbf{Z} = \{\mathbf{z}_n\}_{n=1}^{T}$ and $\Theta = \{\theta_\mathrm{s}, \theta_\mathrm{d}, \psi\}$, the factorization is

$$p_\Theta(\mathbf{Z} \mid x) = \prod_{n=1}^{T} \Bigg[p_{\theta_\mathrm{s}}\left(z_n^{(1)} \mid \mathbf{s}_n\right) \times \prod_{k=2}^{K} p_{\theta_\mathrm{d}}\left(z_n^{(k)} \mid \mathbf{r}_n, \mathbf{z}_n^{(<k)}\right) \Bigg]. \tag{1}$$

The first factor is time-NAR: neither generated semantic history $\mathbf{z}_{<n}^{(1)}$ nor residual history $\mathbf{c}_n$ enters the prediction of $z_n^{(1)}$. The second factor is intra-frame depth-AR: it preserves the coarse-to-fine RVQ order, while cross-frame generated-token recurrence remains confined to residual acoustics. All K tokens share frame index n; depth recursion therefore adds serial computation but no input-frame look-ahead. The codec decoder renders each completed tuple, so inference emits one 26.67-ms codec frame per call (Fig. 1).

2.2. Semantic Codec and Time-NAR Prediction

Following Fish Audio S2 [16], we adapt a causal semantic RVQ codec to 24-kHz speech. A 320-sample internal hop followed by $2\times$ temporal downsampling yields the 37.5-Hz frame rate; each of the ten codebooks has 1024 entries. To encourage CB_1 to serve as a content anchor, codec pretraining aligns its quantized embedding with time-aligned features from a frozen w2v-BERT 2.0 teacher after a learned projection matches their dimension and frame rate [17]. This feature-level distillation is used only during codec pretraining; the teacher is absent from enhancement training and inference.

For enhancement, a 1024-point STFT (320-sample hop, Hann window) and a causal stride-two convolutional encoder produce codec-aligned features $\{\mathbf{u}_i\}$ from degraded speech. For frame n, a Qwen2-0.5B-initialized backbone reads the W features ending at $n + N_{\mathrm{LA}}$ and outputs $\mathbf{s}_n$, where N_{LA} counts strictly future codec frames. It uses causal sliding-window attention with ALiBi [18, 19]; the input–target shift supplies all look-ahead. Crucially, generated CB_1 tokens are never appended to the backbone input, so its KV cache contains degraded-speech evidence only. This absence of temporal target-token feedback is precisely what time-NAR denotes.

2.3. Intra-frame Depth-AR Residual Reconstruction

The residual predictor follows the slow-time/fast-depth organization of Fish-Speech and Fish Audio S2 [20, 16]. Two causally masked Transformer layers (width 896, eight heads, $4H$ feed-forward dimension) condition every depth step on $\mathbf{r}_n$ while exposing only the same-frame prefix $\mathbf{z}_n^{(<k)}$. Thus, CB_2–CB_{10} are generated sequentially in RVQ order, but the nine depth steps require no additional input frames.

After each frame, a single-layer 896-dimensional GRU consumes the mean of codebook-position-aware embeddings of the residual tokens. Its layer-normalized output, scaled by a learned sigmoid gate (initialized with $\beta = -2$), forms $\mathbf{c}_{n+1}$; we set $\mathbf{c}_1 = \mathbf{0}$. The update uses clean residual tokens and 0.2 history-context dropout during training, and generated tokens during inference. Because $\mathbf{c}_n$ enters only $\mathbf{r}_n$, this recurrence provides causal acoustic context without reopening the semantic feedback path.

After codec pretraining, we optimize codebook-weighted cross-entropy over the factors in Eq. (1). Teacher forcing supplies clean same-frame prefixes and clean residual history; inference replaces both with generated tokens.

3. EXPERIMENTS

3.1. Experimental Setup

Edge-GSE reuses the DelayGSE training corpus and mixture simulator [21], comprising 30,292 h of clean speech, approximately 590 h of noise, and 700,000 simulated reverberant examples. Every variant is trained for 300,000 steps with Adam, a peak learning rate of 5×10^{-4}, 5,000-step warm-up followed by cosine decay, and dynamic batches of approximately 40 s.

We evaluate on 400 utterances from URGENT 2025 `nonblind_test` [15]: 300 English and 100 Chinese utterances. We report DNSMOS, cosine speaker similarity (SIM), and the equal-language error $\mathrm{Macro\,ASR} = 100(\mathrm{WER}_{\mathrm{EN}} + \mathrm{CER}_{\mathrm{ZH}})/2$. The Chinese term remains CER; this metric is not a pooled word-level rate. Aggregate objective results are fixed-checkpoint point estimates.

All newly evaluated systems share the same manifest and scoring pipeline. We use released checkpoints for UniSE, Stream.FM, FastEnhancer-L, and DeepFilterNet3; Bridge is retrained from public code on internal data [22], while DelayGSE is an inherited internal record. Baseline training data and latency definitions are not harmonized. The streaming rows therefore provide a broadly comparable interactive reference, while the offline rows provide global context. Only the ablations match the manifest, evaluators, training recipe, and effective context budget.

For the depth-factorization comparison at $N_{\mathrm{LA}} = 6$, eight listeners provide 160 five-level, double-blind paired judgments over 119 unique clips for parallel versus intra-frame depth-AR residual prediction. Presentation order is varied, and preference strengths are collapsed for headline reporting.

3.2. Q1: Comparison at the Primary Operating Point

At the primary $N_{\mathrm{LA}} = 6$ point, Edge-GSE obtains 3.62 DNSMOS, 25.45% Macro ASR, and 0.708 SIM, giving the best point estimate in all three columns of the streaming block. This is an interactive-regime comparison, not an exact end-to-end latency match: Edge-GSE uses 186.7-ms input context, whereas external systems retain their reported configurations and delay definitions.

Discrete-token enhancement may need more acoustic evidence before categorical semantic decisions stabilize. The lower Macro ASR at larger contexts in Fig. 2 is consistent with this hypothesis, so the stronger recognition result must be read together with its larger delay.

The offline rows supply a global view: UniSE has higher DNSMOS, while DelayGSE has higher DNSMOS and lower Macro ASR but lower SIM. Noisy input retains the lowest Macro ASR (15.55%), so this fixed-backend metric supports controlled comparison rather than absolute transcript-preservation claims.

3.3. Q2: Factorization and Scheduling Controls

All diagnostics use the primary 186.7-ms effective input-context budget. The first row of Table 2 is a delayed-grid scheduling control; the four frame-aligned rows form the matched cumulative ablation. All share the evaluation and training recipe, but the scheduling row does not isolate temporal feedback.

The two time-AR controls retain parallel residual prediction and no history; only their target schedules differ. Frame

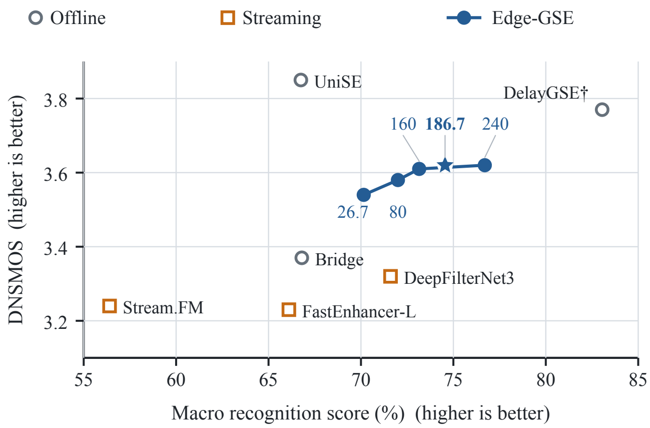

Fig. 2. DNSMOS–recognition trade-off on URGENT 2025 (upper right is better). Macro recognition score is $100-$Macro ASR error—a direction-aligned score, not token accuracy. Blue labels give input context (ms); the star marks the 186.7-ms primary record. Noisy input is omitted; † denotes an inherited record.

alignment gives mixed changes (3.50 to 3.48 DNSMOS, 48.80% to 48.40% Macro ASR, and 0.588 to 0.504 SIM), so the delayed-grid row is a scheduling reference rather than temporal-feedback evidence.

The frame-aligned time-AR and time-NAR rows isolate temporal generated-token feedback. With target alignment, residual decoding, and recurrence fixed, removing it reduces Macro ASR from 48.40% to 26.60% (45.0% relative), raises DNSMOS from 3.48 to 3.58, and raises SIM from 0.504 to 0.677—the largest matched effect.

With time-NAR fixed, depth-AR changes objective scores modestly (3.58 to 3.57 DNSMOS, 26.60% to 26.05% Macro ASR, and 0.677 to 0.696 SIM), but the listening test yields 103 preferences for depth-AR, 42 ties, and 15 for parallel prediction. The direction holds in both display positions and for seven of eight listeners, supporting coarse-to-fine conditioning perceptually.

Residual-only recurrence further improves DNSMOS by 0.05, Macro ASR by 0.60 points, and SIM by 0.012 without reopening the semantic feedback path.

3.4. Q3: Context–Quality Trade-off and Device Latency

For codec-frame duration $\Delta = 26.67$ ms, the context-plus-processing latency is $L_{\mathrm{A2A}} = (N_{\mathrm{LA}}+1)\Delta + \mathrm{RTF}\,\Delta$. At the primary $N_{\mathrm{LA}} = 6$ point, 186.7-ms input context and 21.60-ms average processing at RTF 0.81 give 208.3 ms. Fig. 2 marks this primary record, shared by Tables 1 and 2; it is separate from the connected sweep.

Along the connected sweep, increasing input context from 26.67 to 240 ms raises DNSMOS from 3.54 to 3.62 and reduces Macro ASR from 29.85% to 23.30%, while SIM remains within 0.711–0.716. At the primary setting, W8A16

Table 1. Comparison on URGENT 2025. Edge-GSE uses the primary 186.7-ms input context; streaming rows use their respective low-latency configurations, and offline rows provide global context. Bold/underlined entries mark the best/second-best streaming results; † denotes an inherited record.

Method	DNSMOS ↑	Macro ASR (%) ↓	SIM ↑
Noisy input	2.74	15.55	0.697
Offline enhancement			
UniSE [23]	3.85	33.25	0.660
DelayGSE† [21]	3.77	16.95	0.634
Bridge [22]	3.37	33.20	0.593
Streaming enhancement			
Stream.FM [10]	3.24	43.60	0.660
FastEnhancer-L [24]	3.23	33.90	0.658
DeepFilterNet3 [25]	<u>3.32</u>	<u>28.40</u>	<u>0.699</u>
Edge-GSE	**3.62**	**25.45**	**0.708**

Table 2. Factorization study at a 186.7-ms context budget. The first (†) row changes to delayed-grid targets; the four frame-aligned rows remove temporal feedback, then add depth-AR and residual recurrence. Bold marks the best result.

Variant	DNSMOS ↑	Macro ASR (%) ↓	SIM ↑
Time-AR + delayed-grid†	3.50	48.80	0.588
Time-AR + parallel residuals	3.48	48.40	0.504
Time-NAR + parallel residuals	3.58	26.60	0.677
Time-NAR + depth-AR	3.57	26.05	0.696
Edge-GSE (+ residual GRU)	**3.62**	**25.45**	**0.708**

C++ reaches RTF 0.81 on MTK 8678; the Qwen2, depth, and GRU modules contain 357.90M, 19.29M, and 4.82M parameters (382.01M total).

4. CONCLUSION

We presented Edge-GSE, which removes temporal feedback from the semantic codebook while retaining intra-frame depth-AR and residual-only history. At 186.7-ms input context, it obtains 3.62 DNSMOS, 25.45% Macro ASR, and 0.708 SIM; matched controls identify temporal-feedback removal as the largest gain, while listener preferences support depth-AR. With 21.60-ms average processing, this point gives 208.3-ms context-plus-processing latency on MTK 8678 at RTF 0.81.

Evidence is limited to 400 bilingual utterances, fixed-checkpoint estimates, and a small descriptive listening test, leaving broader conditions and paired uncertainty unmeasured. Baseline latency definitions are not harmonized, and the device result reports average processing on one platform rather than cross-device tail latency. Future work will combine multilingual, content-focused evaluation with adaptive context and token prediction to preserve semantic stability at shorter delay.

5. REFERENCES

[1] Yong Xu, Jun Du, Li-Rong Dai, and Chin-Hui Lee, "A regression approach to speech enhancement based on deep neural networks," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 23, no. 1, pp. 7–19, 2015.

[2] Yanxin Hu, Yun Liu, Shubo Lv, Mengtao Xing, Shimin Zhang, Yihui Fu, Jian Wu, Bihong Zhang, and Lei Xie, "DCCRN: Deep complex convolution recurrent network for phase-aware speech enhancement," in *Interspeech 2020*, 2020, pp. 2472–2476.

[3] Xiang Hao, Xiangdong Su, Radu Horaud, and Xiaofei Li, "FullSubNet: A full-band and sub-band fusion model for real-time single-channel speech enhancement," in *ICASSP 2021 – 2021 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2021, pp. 6633–6637.

[4] Jean-Marie Lemercier, Julius Richter, Simon Welker, and Timo Gerkmann, "Storm: A diffusion-based stochastic regeneration model for speech enhancement and dereverberation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 2724–2737, 2023.

[5] Ziqian Wang, Zikai Liu, Xinfa Zhu, Yike Zhu, Mingshuai Liu, Jun Chen, Longshuai Xiao, Chao Weng, and Lei Xie, "FlowSE: Efficient and High-Quality Speech Enhancement via Flow Matching," in *Interspeech*, 2025, pp. 4858–4862.

[6] Ante Jukić, Roman Korostik, Jagadeesh Balam, and Boris Ginsburg, "Schrödinger bridge for generative speech enhancement," in *Interspeech 2024*, 2024, pp. 1175–1179.

[7] Ziqian Wang, Xinfa Zhu, Zihan Zhang, YuanJun Lv, Ning Jiang, Guoqing Zhao, and Lei Xie, "Selm: Speech enhancement using discrete tokens and language models," in *IEEE ICASSP*, 2024, pp. 11561–11565.

[8] Haici Yang, Jiaqi Su, Minje Kim, and Zeyu Jin, "Genhancer: High-fidelity speech enhancement via generative modeling on discrete codec tokens," in *Interspeech*, 2024, pp. 1170–1174.

[9] Jixun Yao, Hexin Liu, Chen Chen, Yuchen Hu, Eng Siong Chng, and Lei Xie, "GenSE: Generative speech enhancement via language models using hierarchical modeling," in *International Conference on Learning Representations*, 2025.

[10] Simon Welker, Bunlong Lay, Maris Hillemann, Tal Peer, and Timo Gerkmann, "Real-time streamable generative speech restoration with flow matching," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 34, pp. 3087–3101, 2026.

[11] Huaying Xue, Xiulian Peng, and Yan Lu, "Low-latency speech enhancement via speech token generation," in *IEEE ICASSP*, 2024, pp. 661–665.

[12] Jingyi Li, Luca Della Libera, Mirco Ravanelli, Mingkun Xu, and Cem Subakan, "LL-SDR: Low-latency speech enhancement through discrete representations," 2026.

[13] Fei Liu and Yang Ai, "ParaGSE: Parallel generative speech enhancement with group-vector-quantization-based neural speech codec," 2026, Accepted to ICASSP 2026.

[14] Sofiene Kammoun, Xavier Alameda-Pineda, and Simon Leglaive, "Modeling strategies for speech enhancement in the latent space of a neural audio codec," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.

[15] Kohei Saijo and Wangyou Zhang and Samuele Cornell and Robin Scheibler and Chenda Li and others, "Interspeech 2025 URGENT Speech Enhancement Challenge," in *Interspeech*, 2025, pp. 858–862.

[16] Shijia Liao, Yuxuan Wang, Songting Liu, Yifan Cheng, Ruoyi Zhang, Tianyu Li, Shidong Li, Yisheng Zheng, Xingwei Liu, Qingzheng Wang, Zhizhuo Zhou, Jiahua Liu, Xin Chen, and Dawei Han, "Fish audio S2 technical report," 2026.

[17] Seamless Communication et al., "Seamless: Multilingual expressive and streaming speech translation," 2023.

[18] An Yang et al., "Qwen2 technical report," 2024.

[19] Ofir Press, Noah A. Smith, and Mike Lewis, "Train short, test long: Attention with linear biases enables input length extrapolation," in *International Conference on Learning Representations*, 2022.

[20] Shijia Liao, Yuxuan Wang, Tianyu Li, Yifan Cheng, Ruoyi Zhang, Rongzhi Zhou, and Yijin Xing, "Fish-Speech: Leveraging large language models for advanced multilingual text-to-speech synthesis," 2024.

[21] Xin Yuan, Junling Lv, Zezhou Xu, Xingjun Tan, Liangliang Li, and Yanqiang Lei, "DelayGSE: A generative speech enhancement framework with delayed text-aware conditioning," Accepted to Interspeech 2026; official proceedings metadata pending, 2026.

[22] Dahan Wang, Jun Gao, Tong Lei, Yuxiang Hu, Changbao Zhu, Kai Chen, and Jing Lu, "Rethinking flow and diffusion bridge models for speech enhancement," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 39, pp. 33431–33439, 2026.

[23] Haoyin Yan, Chengwei Liu, Shaofei Xue, Xiaotao Liang, Yinghao Liu, Yuxiang Kong, and Zheng Xue, "UniSE: A unified framework for decoder-only autoregressive LM-based speech enhancement," 2025, Accepted to Interspeech 2026.

[24] Sunghwan Ahn, Jinmo Han, Beom Jun Woo, and Nam Soo Kim, "FastEnhancer: Speed-optimized streaming neural speech enhancement," in *ICASSP 2026 – 2026 IEEE International Conference on Acoustics, Speech and Signal Processing*. May 2026, IEEE.

[25] Hendrik Schröter, Alberto N. Escalante-B., Tobias Rosenkranz, and Andreas Maier, "DeepFilterNet: Perceptually motivated real-time speech enhancement," in *Interspeech 2023*, 2023, pp. 2008–2009.